Macroeconomics and household heterogeneity: algorithms

1 Markov Chains

A discrete-time Markov chain featuring a discrete and finite state space $Y = \{y_1, y_2, ..., y_M\}$ is a stochastic process $\{\hat{y}_t\}_{t\geq 0}$ (i.e. a sequece of random variables) satisfying the Markov property:

$$\mathbb{P}\left(\hat{y}_t | \hat{y}_0, \hat{y}_1, ..., \hat{y}_{t-1}\right) = \mathbb{P}\left(\hat{y}_t | \hat{y}_{t-1}\right)$$

i.e., the probability of moving to the next-period state depends solely on the current state.

Markov chain is characterized by:

1. State space $Y = \{y_1, y_2, ..., y_M\}$
2. Transition matrix $\pi = \{\pi_{jk}\}_{j,k\in\{1,2,...,M\}} = \{\mathbb{P}\left(y_k | y_j\right)\}_{j,k\in\{1,2,...,M\}}$ specifying the transition probability of moving from state y_j to state y_k for all $k,\ j \in \{1, 2, ..., M\}$, where:

$$\forall_{j\in\{1,2,...,M\}} \sum_{k=1}^{M} \mathbb{P}\left(y_k | y_j\right) = 1$$

3. Initial distribution $\mathbb{P}\left(\hat{y}_0\right)$ where $\hat{y}_0 \in Y$ and

$$\sum_{k=1}^{M} \mathbb{P}\left(\hat{y}_0 = y_k\right) = 1$$

A stationary distribution μ is a $1 \times M$ vector such that:

$$\mu = \mu \cdot \pi$$

We say that states y_k and y_j **communicate with each other** if the entries jk and kj of matrices π^{n_1} and π^{n_2} are strictly positive for some n_1 and n_2. State y_j is **periodic** if there exists a number $J > 1$ such that J is the greatest common divisor of the set:

$$\left\{n :\ \pi_{jj}^{n} > 0\right\}.$$

If such a divisor is $J = 1$ then y_j is aperiodic.

From the probability theory we have the following fact (theorem): if the Markov chain is irreducible and aperiodic then there is a unique stationary distribution μ. Moreover, for all k and j we have that the entry jk of matrix π^n coverges to μ_k for $n \to +\infty$ (i.e. starting from an arbitrary distribution $\mathbb{P}(\hat{y}_0)$ the iterated distribution converges to the stationary one). This provides the first way of computing the stationary distribution.

Morover, notice that

$$\mu = \mu \cdot \pi \Longleftrightarrow \pi^T \cdot \mu^T = \mu^T$$

$$\Longleftrightarrow \pi^T \cdot \mu^T = 1 \cdot \mu^T$$

and thus μ^T can be thought of as the eigenvector that corresponds to eigenvalue equal to unity. This provides the second way of computing the stationary distribution.

2 Discretization of the AR(1) process

Suppose that individual income is given by $\exp(y_t)$ where its logarithm evolves according to:

$$y_t = \rho \cdot y_{t-1} + \epsilon_t$$

where $\rho \in (-1, 1)$ and $\epsilon_t \in N\left(0, \sigma^2\right)$ is i.i.d. over time. Now:

$$\mathbb{E}y_t = \mathbb{E}\left(\epsilon_t + \rho \cdot \epsilon_{t-1} + \rho^2 \cdot \epsilon_{t-2} + ...\right)$$

$$= \mathbb{E}\epsilon_t + \rho \cdot \mathbb{E}\epsilon_{t-1} + \rho^2 \cdot \mathbb{E}\epsilon_{t-2} + ... = 0.$$

$$Var\left(y_t\right) = Var\left(\epsilon_t + \rho \cdot \epsilon_{t-1} + \rho^2 \cdot \epsilon_{t-2} + ...\right)$$

$$\overset{\text{i.i.d.}}{=} Var\left(\epsilon_t\right) + \rho^2 \cdot Var\left(\epsilon_t\right) + \rho^4 \cdot Var\left(\epsilon_t\right) + ...$$

$$= \frac{Var\left(\epsilon_t\right)}{1 - \rho^2} = \frac{\sigma^2}{1 - \rho^2}.$$

We will now discuss two widely-used methods of discretizing the process $\{y_t\}_{t \geq 0}$.

2.1 Tauchen

This algorithm is by Tauchen [1986]. Set:

$$y_M = m \cdot \left(\frac{\sigma^2}{1 - \rho^2}\right)^{\frac{1}{2}}$$

and:

$$y_1 = -m \cdot \left(\frac{\sigma^2}{1 - \rho^2}\right)^{\frac{1}{2}}$$

where m is usually equal to 3.

Let us now create a grid of equidistant points $[y_1, y_2, ..., y_M]$ with the distance equal to D. Let us construct the transition matrix for $k \in \{2, 3, ..., M-1\}$:

$$\pi_{jk} = \mathbb{P}(y_k|y_j) = \mathbb{P}\left(y_k - \frac{D}{2} \leq \rho \cdot y_j + \epsilon \leq y_k + \frac{D}{2}\right)$$

$$= \mathbb{P}\left(y_k - \frac{D}{2} - \rho \cdot y_j \leq \epsilon \leq y_k + \frac{D}{2} - \rho \cdot y_j\right)$$

$$= \Phi\left(\frac{y_k + \frac{D}{2} - \rho \cdot y_j}{\sigma}\right) - \Phi\left(\frac{y_k - \frac{D}{2} - \rho \cdot y_j}{\sigma}\right)$$

where Φ is the cdf of $N(0,1)$. Moreover:

$$\pi_{jM} = 1 - \Phi\left(\frac{y_M - \frac{D}{2} - \rho \cdot y_j}{\sigma}\right)$$

$$\pi_{j1} = \Phi\left(\frac{y_1 + \frac{D}{2} - \rho \cdot y_j}{\sigma}\right).$$

Exercise: write a code implementing the algorithm and simulate the path of a stochastic realization corresponding to that Markov chain (given the starting point and the path length).

2.2 Rouwenhorst

This discretization method is better than the previous one for the approximation of persistent AR(1) processes that are common in macroeconomics. It is based on the Rouwenhorst's chapter in Cooley and Prescott [1995].

We set $-y_1 = y_M = \omega$. Now, let us build the transition matrix π in the following recursive way:

- If $M = 2$ then:
$$\pi(M) = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$
- If $M > 3$ then:
$$\pi(M) = p \cdot \begin{bmatrix} \pi(M-1) & \mathbf{0} \\ \mathbf{0^T} & 0 \end{bmatrix} + (1-p) \cdot \begin{bmatrix} \mathbf{0} & \pi(M-1) \\ 0 & \mathbf{0^T} \end{bmatrix}$$
$$+ (1-p) \cdot \begin{bmatrix} \mathbf{0^T} & 0 \\ \pi(M-1) & \mathbf{0} \end{bmatrix} + p \cdot \begin{bmatrix} 0 & \mathbf{0^T} \\ \mathbf{0} & \pi(M-1) \end{bmatrix}$$
where $\mathbf{0}$ is a $(M-1) \times 1$ vector of zeros.
- it can be checked that the elements in all but the top and bottom rows add up to 2 so they have to be standardized by dividing them by 2

This method works because the invariant distribution $\mu(M)$ of the Markov chain with the transition matrix $\pi(M)$ is:

$$\mu_j(M) = \begin{pmatrix} M-1 \\ j-1 \end{pmatrix} \cdot \left(\frac{1}{2}\right)^{M-j-2}$$
$$= \begin{pmatrix} M-1 \\ j-1 \end{pmatrix} \cdot \left(\frac{1}{2}\right)^{M-1} \cdot \left(\frac{1}{2}\right)^{j-1}$$

which is the chance of obtaining $j-1$ successes in $M-1$ independent Bernoulli trials featuring the success probability of 0.5. The moments of that distribution are (see Kopecky and Suen [2010]):

$$\mathbb{E}y_t = 0$$
$$Var(y_t) = \frac{1}{M-1} \cdot \omega^2$$

which gives us a simple way of computing ω by equating this expression with the variance of the AR(1) process:

$$\frac{\sigma^2}{1-\rho^2} = \frac{1}{M-1} \cdot \omega^2$$
$$\Longrightarrow \omega = \sigma \cdot \sqrt{\frac{M-1}{1-\rho^2}}.$$
$$p = \frac{1+\rho}{2}$$

Moreover, using the fact that as $M \to +\infty$ the binominal distribution $\mu(M)$ converges to the Normal distribution we see that the proposed approximation is adequate for the AR(1) process with the Gaussian standard error.

Exercise: write a code implementing the algorithm.

3 Bisection

Let $g(x)$ be a strictly monotonic function (WLOG strictly increasing) and suppose that there is an interval $[\underline{x}, \bar{x}]$ such that:

$$g(\underline{x}) < 0 < g(\bar{x}).$$

We want to find x^* s.t. $g(x) = 0$. We have the following procedure:

1. Set $x_L = \underline{x}$, $x_R = \bar{x}$.
2. Compute $x_M = \frac{x_L + x_R}{2}$
3. If $g(x_M) > 0$ then we set $x_R = x_M$, if $g(x_M) < 0$ we set $x_L = x_M$ and go back to step 2.

Continue the procedure until both $|g(x_L) - g(x_R)|$ and $|x_L - x_R|$ are sufficiently small.

4 Value Function Iterations and the Endogenous Grid Method

4.1 Maximization problem

Let us consider the following Bellman equation:

$$V(a,y) = \max_{c,\, a'} \left\{ u(c) + \beta \mathbb{E}_{y'|y} V(a', y') \right\}$$

$$\text{subject to:}$$

$$c + a' = R \cdot a + y$$

$$a' \geq -\phi$$

where V is the value function, a is the level of assets at the beginning of the current period, c is consumption, y is exogenous (stochastic) income, β is a discount factor and $R < \frac{1}{\beta}$ is the real interest rate. Evolution of y is governed by transition matrix $\mathbb{P}(y'|y)$. Importantly, current asset choice a' is constrained by lower bound $-\phi$ where $\phi > 0$ is a parameter. In addition to V, there are two policy functions $a'(a,y)$ and $c(a,y)$ describing the solution to the maximization problem.

The Euler inequality (note that borrowing is constrained) can be calculated by combining the Envelope condition with the FOC is given by:

$$u'(c) \geq \beta \cdot R \cdot \mathbb{E}_{y'|y} u'(c').$$

Note that if the optimal policy is $a' = -\phi$ then $MPC = 1$ which has some important consequences for the behavior of aggregate demand if the distribution of agents is concentrated in the neighborhood of $-\phi$.

4.2 Value Function Iterations (VFI)

Let us define the grid $A = \{a_1, a_2, ..., a_N\}$ where $a_1 = -\phi$ and the grid for income $Y = \{y_1, y_2, ..., y_M\}$. Start by value function $V^i(a,y)$ on $A \times Y$. Solve the problem:

$$V^{i+1}(a,y) = \max_{a'} \left\{ u(c) + \beta \mathbb{E}_{y'|y} V^i(a', y') \right\}$$

$$\text{subject to:}$$

$$c + a' = R \cdot a + y$$

$$a' \geq -\phi$$

to get $V^{i+1}(a,y)$. Check for convergence, i.e., whether:

$$\left\| V^i - V^{i+1} \right\| < \epsilon$$

where ϵ is a small positive number (convergence criterion) and $\|\cdot\|$ is a norm defined on the space of functions (e.g. the max norm, the L^1 norm, etc.). If it

is satisfied then stop the algorithm - the value function is found. If not then substitute:

$$V^i(a,y) := V^{i+1}(a,y)$$

and repeat the step above.

The logic behind the VFI algorithm relies on the application of the Banach Theorem. More specifically, we define an operator T on the metric space $(\mathcal{V}, \|\cdot\|)$ of functions with the norm $\|\cdot\|$:

$$T(V) \equiv \max_{a'} \left\{ u(c) + \beta \mathbb{E}_{y'|y} V(a', y') \right\}$$

$$\text{subject to:}$$

$$c + a' = R \cdot a + y$$

$$a' \geq -\phi.$$

One can use the Blackwell Theorem to show that T is a contraction mapping, i.e., there exists a number $\delta \in (0,1)$ such that for any functions V^j and V^k from the metric space $(\mathcal{V}, \|\cdot\|)$:

$$\delta \cdot \left\| V^j - V^k \right\| > \left\| T\left(V^j\right) - T\left(V^k\right) \right\|.$$

Now, if T is a contraction mappin then the Banach Theorem implies that there exists a unique solution (fixed point) to equation:

$$T(V) = V$$

and it can be found using an iterative procedure that starts from an arbitrary element V^0 of space $(\mathcal{V}, \|\cdot\|)$:

$$V^0, T\left(V^0\right), T\left(T\left(V^0\right)\right), ... \to V$$

4.3 The EGM algorithm

Let us define the grid $A = \{a_1, a_2, ..., a_N\}$ where $a_1 = -\phi$ and the grid for income $Y = \{y_1, y_2, ..., y_M\}$. Start by guessing consumption policy $c^0(a,y)$ on $A \times Y$.

1. Use the Euler equation (the "inequality part" of that condition will be dealt with below) to get the current consumption quasi-policy:

$$\bar{c}(a_i, y_j) = (u')^{-1} \left(\beta \cdot R \cdot \sum_{k=1}^{M} \mathbb{P}(y_k | y_j) \cdot u'\left(c^0(a_i, y_k)\right) \right)$$

   where $\bar{c}(a_i, y_j)$ can be thought of as the consumption policy related to the optimal choice of current assets equal to a_i (so its first argument is not the current state variable, that is why it is referred to as "quasi").

2. Now, use optimal choices $\bar{c}(a_i, y_j)$ and a_i together with the budget constraint to obtain the value of current assets that is consistent with them:

$$\bar{a}_{ij} = \frac{\bar{c}(a_i, y_j) + a_i - y_j}{R}.$$

As a byproduct, we obtain the updated guess for consumption policy $c^1(\bar{a}_{ij}, y_j)$ on endogenous grid $\{\bar{a}_{ij}\}$ and Y.

3. We use $c^1(\bar{a}_{ij}, y_j)$ to pin down the value of the updated guess for $y = y_j$ and $a \in A$:

   (a) If $a_l \in A$ satisfies $a_l < \bar{a}_{1j}$ then we set from the household budget constraint:

   $$c^1(a_l, y_j) = R \cdot a_l + y_j - a_1.$$

   This is because we know that for $\{\bar{a}_{1j}, y_j\}$, the optimal choice of assets is $a_1 = -\phi$ and hence, for $a_l \in A$ satisfying $a_l < \bar{a}_{1j}$ the household wants to borrow up to the limit (i.e., the optimal asset choice from the elements of A is a_1, i.e., the household would like to borrow more but this option is not available due to the limit - it wants to borrow more because its assets a_l are lower than asset level $\bar{a}_{1j}$ for which the optimal saving policy equals a_1).

   (b) If $a_l \in A$ satisfies $a_l > \bar{a}_{Nj}$ then (by analogy to the case when $a_l < \bar{a}_{1j}$) we set from the household budget constraint:

   $$c^1(a_l, y_j) = R \cdot a_l + y_j - a_N.$$

   I.e., the household with current assets exceeding $\bar{a}_{Nj}$ would like to save more than a_N but the available option from A that is closest to their optimum is a_N.

   (c) If $a_l \in A$ satisfies $\bar{a}_{1j} \leq a_l \leq \bar{a}_{Nj}$ then we search for the points from the endogenous grid $\{\bar{a}_{ij}\}$ that are the closest neighbors of a_l: $\bar{a}_{kj} < a_l < \bar{a}_{k+1,j}$. Then we interpolate to get $c^1(a_l, y_j)$:

   $$c^1(a_l, y_j) = \frac{a_l - \bar{a}_{kj}}{\bar{a}_{k+1,j} - \bar{a}_{kj}} \cdot c^1(\bar{a}_{k+1,j}, y_j) + \frac{\bar{a}_{k+1,j} - a_l}{\bar{a}_{k+1,j} - \bar{a}_{kj}} \cdot c^1(\bar{a}_{kj}, y_j).$$

4. We compare $c^1(a_l, y_j)$ and $c^0(a_l, y_j)$ using some notion of distance (e.g., the average absolute difference, the maximum absolute difference). If they are close enough, we stop the algorithm. If they are still distant, we set $c^0(a_l, y_j) = c^1(a_l, y_j)$ and return to point 1.

5 The Huggett model

5.1 Description of the model

The maximization problem of households is described in subsection 4.1. To define the equilibrium in the model, we have to specify the distribution of agents

over states a and y over time and define the market clearing conditons. From now on, it is assumed that $\phi > 0$ (i.e. credit is allowed) and R becomes endogenous (it clears the market for assets; the price of consumption goods is standardized to unity and the market for goods clears from the Walras law).

The evolution of the distribution of agents $\mu(a, y)$ (distribution over assets and income/productivity levels) over time is described by:

$$\mu'(\bar{a}', \bar{y}') = \sum_{a \in A, y \in Y} \mathbb{P}(\bar{y}'|y) \cdot \mathbb{I}_{\{a'(a,y)=\bar{a}'\}} \cdot \mu(a, y) \tag{1}$$

i.e., those, who have assets $\bar{a}'$ and $\bar{y}'$ in the next period are those, who choose assets $\bar{a}'$ today and their productivity becomes $\bar{y}'$ in the future. Note that $\mathbb{I}$ is the indicator function. Moreover, we standardize:

$$\sum_{a \in A, y \in Y} \mu(a, y) = 1.$$

Finally, the market clearing conditions (for goods and assets, respectively) are:

$$\sum_{a \in A, y \in Y} c(a, y) \cdot \mu(a, y) = \sum_{a \in A, y \in Y} y \cdot \mu(a, y) \tag{2}$$

$$\sum_{a \in A, y \in Y} a'(a, y) \cdot \mu(a, y) = 0. \tag{3}$$

The latter implies that, in the equilibrium, some agents are savers and, effectively, lend funds to those who are indebted.

We define the equilibrium:

Definition (equilibrium in the Huggett model) Equilibrium in the Huggett model are: interest rate R, value function $V(a, y)$, policy functions $a'(a, y)$, $c(a, y)$ and the stationary measure μ^* such that:

1. Given R and $\mathbb{P}(y'|y)$, value function $V(a, y)$ and policy functions $a'(a, y)$, $c(a, y)$ solve the household's problem
2. Market clearing conditions for goods and assets hold
3. Measure μ^* satisfies condtion 1 (is the fixed point of that condition $\mu = \mu' = \mu^*$)

5.2 Solution algorithm - stationary equilibrium

To find the equilibrium in the Huggett model, we proceed as follows. First, we guess R^0 and use the same exogenous grids as in subsection 4.3. We also set tolerance level for the accuracy of the solution $\epsilon > 0$. Then we take the following steps:

1. Given R^0 we compute function $c(a, y)$ using the EGM algorithm (see subsection 4.3).

2. Given $c(a, y)$ we compute the asset policy function:

$$a'(a_i, y_j) = R^0 \cdot a_i + y_j - c(a_i, y_j)$$

for all $a_i \in A$ and $y_j \in Y$.

3. Using $a'(a, y)$ and $\mathbb{P}(y'|y)$ we compute the transition matrix T for households (defined for all possible asset and income/productivity levels). To this end, we first initialize a $N \cdot M \times N \cdot M$ matrix T of zeros. The order of rows/columns in T corresponds to N vectors of length M (productivity/income) stacked one after another. and then, for all $i \in \{1, 2, ..., N\}$ and $j \in \{1, 2, ..., M\}$, we do the following:

   (a) We search for the points from the exogenous grid A that are the closest neighbors of $a'(a_i, y_j)$: $a_k < a'(a_i, y_j) < a_{k+1}$. Then we interpolate to get weights (equal to the proportions of agents featuring individual states a_i and y_j that have asset level equal to a_k or a_{k+1} in the next period):

$$W_k = \frac{a_{k+1} - a'(a_i, y_j)}{a_{k+1} - a_k}$$

$$W_{k+1} = \frac{a'(a_i, y_j) - a_k}{a_{k+1} - a_k} = 1 - W_k.$$

   (b) We fill in the row of matrix T that corresponds to the current productivity/income y_j and assets a_i (i.e. row number $(i-1) \cdot M + j$) by assigning the $2 \cdot M$ vector (note that $\mathbb{P}(\cdot|y_j)$ is the j-th row of matrix $\mathbb{P}$):

$$[W_k \cdot \mathbb{P}(\cdot|y_j) \;\; W_{k+1} \cdot \mathbb{P}(\cdot|y_j)]$$

   from column $(k-1) \cdot M + 1$ to column $(k + 1 - 1) \cdot M + M$.

4. We compute the stationary distribution associated with the transition matrix T, i.e., a $1 \times N \cdot M$ vector μ^* that satisfies:

$$\mu^* = \mu^* \cdot T$$

which, in fact, captures the law of motion of the distribution of agents described by condition 1.

5. We check the asset market clearing conditon is satisfied (by the Walras Law, the market clearing condition for consumption goods is satisfied then the asset market clearing conditon holds):

$$\left| \sum_{i=1}^{N} \sum_{j=1}^{M} a'(a_i, y_j) \cdot \mu^*(a_i, y_j) - 0 \right| < \epsilon.$$

If it is satisfied, we stop. If not, we turn back to point 1. and decrease R^0 if

$$\sum_{i=1}^{N}\sum_{j=1}^{M} a'\left(a_i, y_j\right) \cdot \mu^*\left(a_i, y_j\right) > 0$$

and we raise it otherwise (a standard algorithm like bisection can be used when searching for the real interest rate that clears the market).

5.3 Solution algorithm: transition dynamics - an example

The idea is to compute the transition between two (the same or different) stationary equilibria. In this example, we compute the transition between two stationary equilibria featuring two different values of β. Suppose that it equals $\beta = \beta_0$ in the initial equilibrium and it is equal to $\beta = \beta_1$ in the final equilibrium. Take a large number $\bar{T}$ - it is the number of periods during which the economy transits from the initial equilibrium to the final one. Solve for the two stationary equilibria and by $c^{\beta_1}(a, y)$ denote the consumption function in the final one. By $\mu^{\beta_0}(a, y)$ denote the distribution of agents in the initial equilibrium and R^{β_0} is the real interest rate in this equilibirum.

Let us set $\bar{T}$ that is sufficiently large (say $\bar{T} = 200$) and let us index the periods of transition with $t \in \left\{1, 2, ..., \bar{T}\right\}$. The convention is that before $t = 1$ the economy is in the initial equilibrium and in $t = 1$ parameter β switches to $\beta = \beta_1$. Let us guess the path of interest rates $\{R_1, ..., R_{\bar{T}}\}$ and let us set (small positive numbers) $\epsilon > 0$ and $\Delta > \epsilon$. Let us also set small $\Omega > 0$. Initialize three arrays (of size $\bar{T} \times N \times M$): $c(t, a, y)$, $\mu(t, a, y)$, and $a'(t, a, y)$. The first array is the transition path of consumption functions, i.e., $c(t, a, y)$ is consumption function in period t. The second is the transition path of distributions of agents, where $\mu(t, a, y)$ is the distribution of agents in period t. The last is the saving policy in period t. Set $c\left(\bar{T}, a, y\right) = c^{\beta_1}(a, y)$ and $\mu(1, a, y) = \mu^{\beta_0}(a, y)$, $R_1 = R^{\beta_0}$. Our goal is to find the equilibrium paths for consumptions, distributions of agents and real interest rates. The following algorithm allows for achieving that:

While $\Delta > \epsilon$:

- BACKWARD ITERATION: for t from $\bar{T} - 1$ downto 1 - use the EGM-like algorithm to compute $c(t, a, y)$ given the guess of the path of real interest rates $\{R_1, ..., R_{\bar{T}}\}$. I.e. in period t, for $a_i \in A$ and $y_j \in Y$.

  1. Use the Euler equation (the "inequality part" of that condition will be dealt with below) to get the current consumption quasi-policy:

     $$\bar{c}(t, a_i, y_j) = \left(u'\right)^{-1}\left(\beta_1 \cdot R_{t+1} \cdot \sum_{k=1}^{M} \mathbb{P}\left(y_k | y_j\right) \cdot u'\left(c\left(t+1, a_i, y_k\right)\right)\right)$$

     (note that for $t = \bar{T} - 1$ we have $c(t+1, a_i, y_k) = c^{\beta_1}(a_i, y_k)$!).

(a) Now, use optimal choices $\bar{c}(t, a_i, y_j)$ and a_i together with the budget constraint to obtain the value of current assets that is consistent with them:

$$\bar{a}_{ij} = \frac{\bar{c}(t, a_i, y_j) + a_i - y_j}{R_t}.$$

As a byproduct, we obtain the updated guess for consumption policy $c(t, \bar{a}_{ij}, y_j)$ on endogenous grid $\{\bar{a}_{ij}\}$ and Y.

(b) We use $c^1(t, \bar{a}_{ij}, y_j)$ to pin down the value of the updated guess for $y = y_j$ and $a \in A$:

i. If $a_l \in A$ satisfies $a_l < \bar{a}_{1j}$ then we set from the household budget constraint:

$$c(t, a_l, y_j) = R_t \cdot a_l + y_j - a_1.$$

ii. If $a_l \in A$ satisfies $a_l > \bar{a}_{Nj}$ then (by analogy to the case when $a_l < \bar{a}_{1j}$) we set from the household budget constraint:

$$c(t, a_l, y_j) = R_t \cdot a_l + y_j - a_N.$$

iii. If $a_l \in A$ satisfies $\bar{a}_{1j} \leq a_l \leq \bar{a}_{Nj}$ then we search for the points from the endogenous grid $\{\bar{a}_{ij}\}$ that are the closest neighbors of a_l: $\bar{a}_{kj} < a_l < \bar{a}_{k+1,j}$. Then we interpolate to get $c(t, a_l, y_j)$:

$$c(t, a_l, y_j) = \frac{a_l - \bar{a}_{kj}}{\bar{a}_{k+1,j} - \bar{a}_{kj}} \cdot c(t, \bar{a}_{k+1,j}, y_j) + \frac{\bar{a}_{k+1,j} - a_l}{\bar{a}_{k+1,j} - \bar{a}_{kj}} \cdot c(t, \bar{a}_{kj}, y_j).$$

(c) We calculate the asset policy associated with $c(t, a_l, y_j)$:

$$a'(t, a_i, y_j) = R_t \cdot a_i + y_j - c(t, a_l, y_j)$$

for all $a_i \in A$ and $y_j \in Y$.

– FORWARD ITERATION: for t from 1 to $\bar{T} - 1$

1. Using $a'(t, a, y)$ and $\mathbb{P}(y'|y)$ we compute the transition matrix T for households (defined for all possible asset and income/productivity levels) in period t. To this end, we first initialize a $N \cdot M \times N \cdot M$ matrix T of zeros. The order of rows/columns in T corresponds to N vectors of length M (productivity/income) stacked one after another. and then, for all $i \in \{1, 2, ..., N\}$ and $j \in \{1, 2, ..., M\}$, we do the following:

(a) We search for the points from the exogenous grid A that are the closest neighbors of $a'(t, a_i, y_j)$: $a_k < a'(t, a_i, y_j) < a_{k+1}$. Then we interpolate to get weights (equal to the proportions of agents featuring individual states a_i and y_j that have asset level equal to a_k or a_{k+1} in the next period):

$$W_k = \frac{a_{k+1} - a'(t, a_i, y_j)}{a_{k+1} - a_k}$$

$$W_{k+1} = \frac{a'(t, a_i, y_j) - a_k}{a_{k+1} - a_k} = 1 - W_k.$$

(b) We fill in the row of matrix T that corresponds to the current productivity/income y_j and assets a_i (i.e. row number $(i-1) \cdot M + j$) by assigning the $2 \cdot M$ vector (note that $\mathbb{P}(\cdot|y_j)$ is the j-th row of matrix $\mathbb{P}$):

$$[W_k \cdot \mathbb{P}(\cdot|y_j) \;\; W_{k+1} \cdot \mathbb{P}(\cdot|y_j)]$$

from column $(k-1) \cdot M + 1$ to column $(k+1-1) \cdot M + M$.

2. We compute the transition of the distribution of agents between periods t and $t+1$:

$$\mu(t+1, \cdot, \cdot) = \mu(t, \cdot, \cdot) \cdot T$$

– CONVERGENCE CHECK:

1. For each $t \in \{1, 2, ..., \bar{T} - 1\}$ compute the excess supply in the asset market in that period

$$\Delta_t = \sum_{i=1}^{N} \sum_{j=1}^{M} a'(a_i, y_j) \cdot \mu(a_i, y_j)$$

2. Compute $\Delta = \max \left| \{\Delta_1, ..., \Delta_{\bar{T}-1}\} \right|$. If $\Delta < \epsilon$ then the equilibirum is found and stop the entire procedure. If not then modify the transition path of real rates as follows. For all $t \in \{1, 2, ..., \bar{T} - 1\}$:

(a) if $\Delta_t > 0$ then decrease R_{t+1} to discourage the agents from saving:

$$R_{t+1} := R_{t+1} - \Omega \cdot \Delta_t$$

(b) if $\Delta_t < 0$ then increase R_{t+1} to encourage the agents to save more:

$$R_{t+1} := R_{t+1} - \Omega \cdot \Delta_t$$

3. Take the modified path of real interest rates $\{R_1, ..., R_{\bar{T}}\}$ (note that R_1 remains unchanged and equals to R^{β_0}) and go back to the BACKWARD ITERATION.

References

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